

Project Title

Skin Disease Detection CNN Project

Skin Disease Detection Using Custom CNN Models

A PyTorch-based image classification project for detecting chickenpox, eczema, and ringworm from skin images. The work compares two custom CNN architectures trained from scratch under coursework constraints: no TensorFlow/Keras, no pretrained models, no transformers, no attention modules, and no ResNet/residual skip connections.

Introduction

Skin Disease Detection CNN Project

Overview

Skin disease recognition from images is a computer vision task where visual patterns such as color, lesion texture, shape, and local contrast are used to classify disease categories. This project builds a supervised image classification pipeline for three skin disease classes: chickenpox, eczema, and ringworm.

Objective

The objective is to train and compare custom CNN models, evaluate them on held-out test data, and analyze both quantitative results and qualitative prediction examples.

Applications (Motivation)

Skin Disease Detection CNN Project

Motivation

A lightweight image classifier can support educational skin-disease screening experiments, demonstrate CNN-based visual feature learning, and provide a reproducible coursework pipeline for preprocessing, training, validation, and evaluation.

Important Limitation

This project is not a medical diagnostic tool. It is intended for academic experimentation only. Real clinical deployment would require expert-labeled data, broader disease coverage, bias analysis, and medical validation.

Dataset Collection Details

Skin Disease Detection CNN Project

Dataset Split and Class Counts

Split	Chickenpox	Eczema	Ringworm	Total
train	137	117	125	379
valid	40	34	36	110
test	20	17	18	55

Total images used in this project: 544. The dataset is organized in final_dataset/train, final_dataset/valid, and final_dataset/test.

Pre-processing (if any)

Skin Disease Detection CNN Project

Image Preparation

Images are loaded as RGB, resized to a square input size, normalized using training-set RGB mean and standard deviation, and converted to PyTorch tensors in channel-first format.

Training Augmentation

During training, images use random resized cropping, horizontal flipping, small rotations, brightness adjustment, contrast adjustment, and color adjustment. Validation and test images are not randomly augmented.

Normalization Values

For the latest Model A run, mean=[0.6852092742919922, 0.5491672158241272, 0.4992219805717468] and std=[0.23969070613384247, 0.22714848816394806, 0.23043037950992584].

Model Training (Methodology)

Skin Disease Detection CNN Project

Frameworks

Training is implemented in PyTorch. Accuracy, precision, recall, F1-score, and confusion matrices are computed with scikit-learn.

Optimization

The models use AdamW optimization, weight decay, label smoothing, inverse-frequency class weights, ReduceLROnPlateau scheduling, and early stopping. Checkpoints are selected using validation macro F1-score so that weaker classes are considered during model selection.

Model A

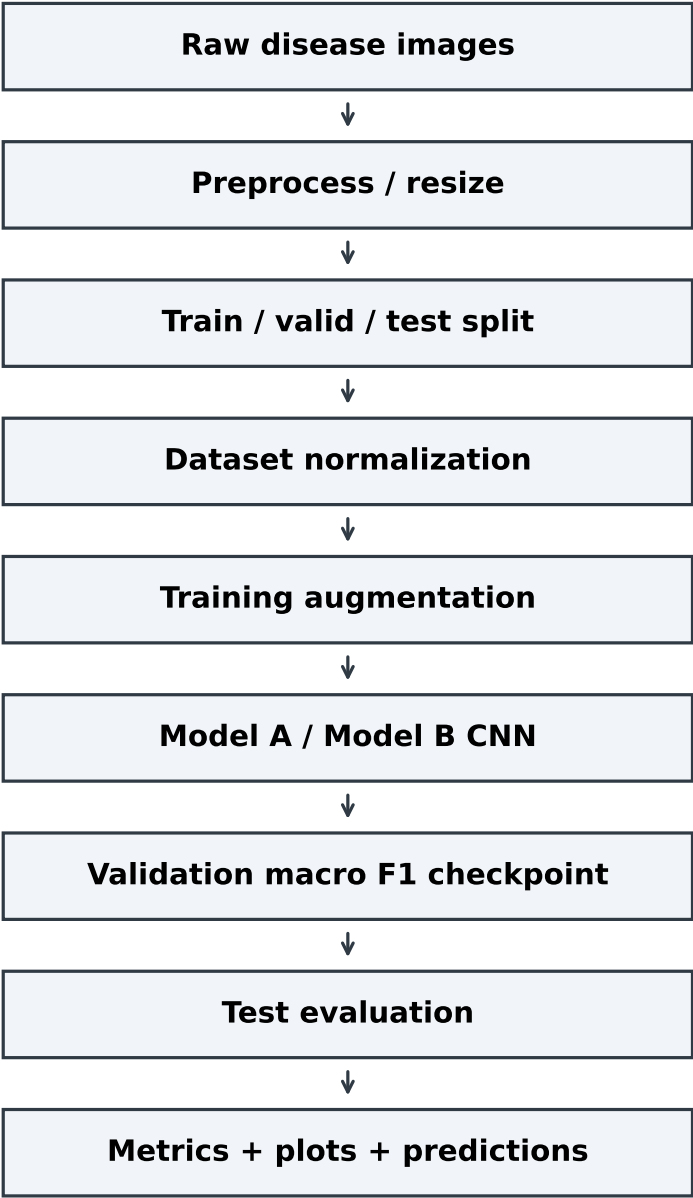
Model A is a VGG-style plain CNN with standard Conv2d blocks, BatchNorm2d, ReLU, MaxPool2d, Dropout2d, and combined AdaptiveAvgPool2d plus AdaptiveMaxPool2d before the classifier.

Model B

Model B is a deeper plain CNN using standard Conv2d stages, BatchNorm2d, SiLU, MaxPool2d, Dropout2d, and combined global average/max pooling. It does not use residual connections or attention.

Complete Pipeline (Figure)

Skin Disease Detection CNN Project



The pipeline is implemented with PyTorch for CNN training and scikit-learn for evaluation metrics.

Experimental Details

Skin Disease Detection CNN Project

Latest Model A Run

Run folder: results/cnn_comparison_20260515_213452. Command: `python -u train_cnn_models.py --models a --image-size 224 --batch-size 8 --epochs 60 --patience 14`.

Latest Model B Run

Run folder: results/cnn_comparison_20260515_200631. This is the latest available completed run that includes Model B metrics.

Evaluation Protocol

The models are evaluated on the held-out test split. Reported accuracy and macro F1 are test-set metrics, while validation loss and validation macro F1 are used for checkpoint selection.

Results (Quantitative / Qualitative)

Skin Disease Detection CNN Project

Quantitative Test Results

Model	Accuracy	Precision	Recall	Macro F1	Val Loss
Model A	69.09%	0.6865	0.6876	0.6831	0.8066
Model B	54.55%	0.5148	0.5303	0.5127	0.8986

Model A is currently the stronger model by test accuracy and macro F1-score.

Results (Quantitative / Qualitative)

Skin Disease Detection CNN Project

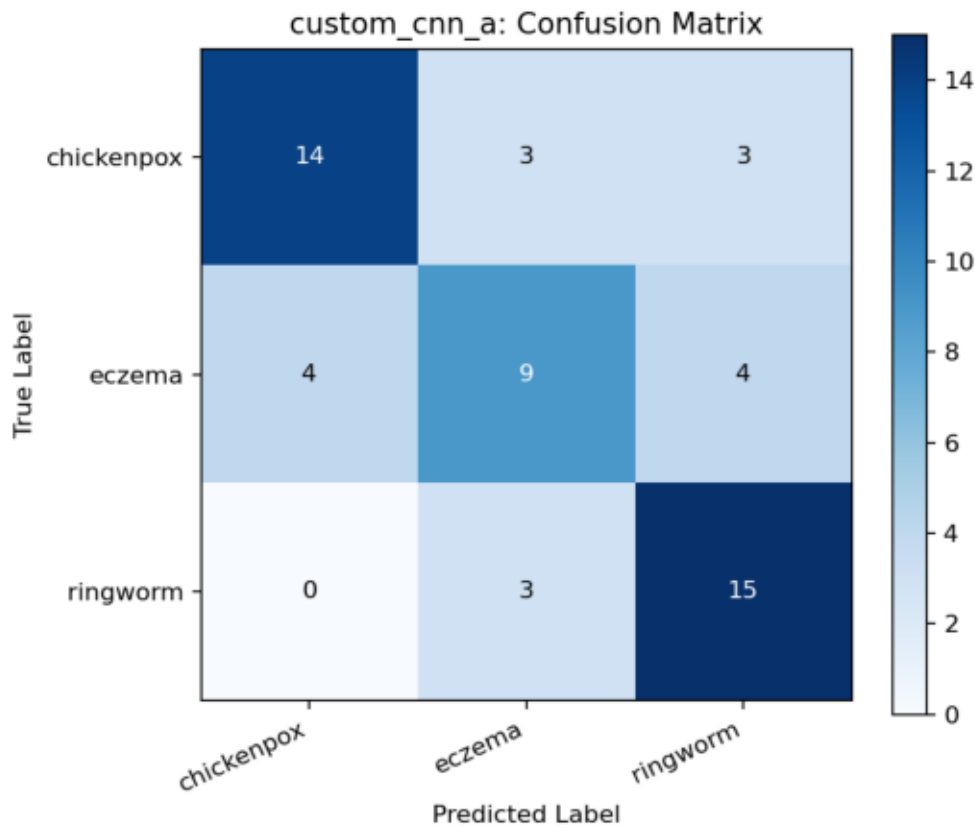
Per-Class Recall and F1-score

Class	A Recall	A F1	B Recall	B F1
chickenpox	0.7000	0.7368	0.8000	0.6809
eczema	0.5294	0.5625	0.2353	0.2857
ringworm	0.8333	0.7500	0.5556	0.5714

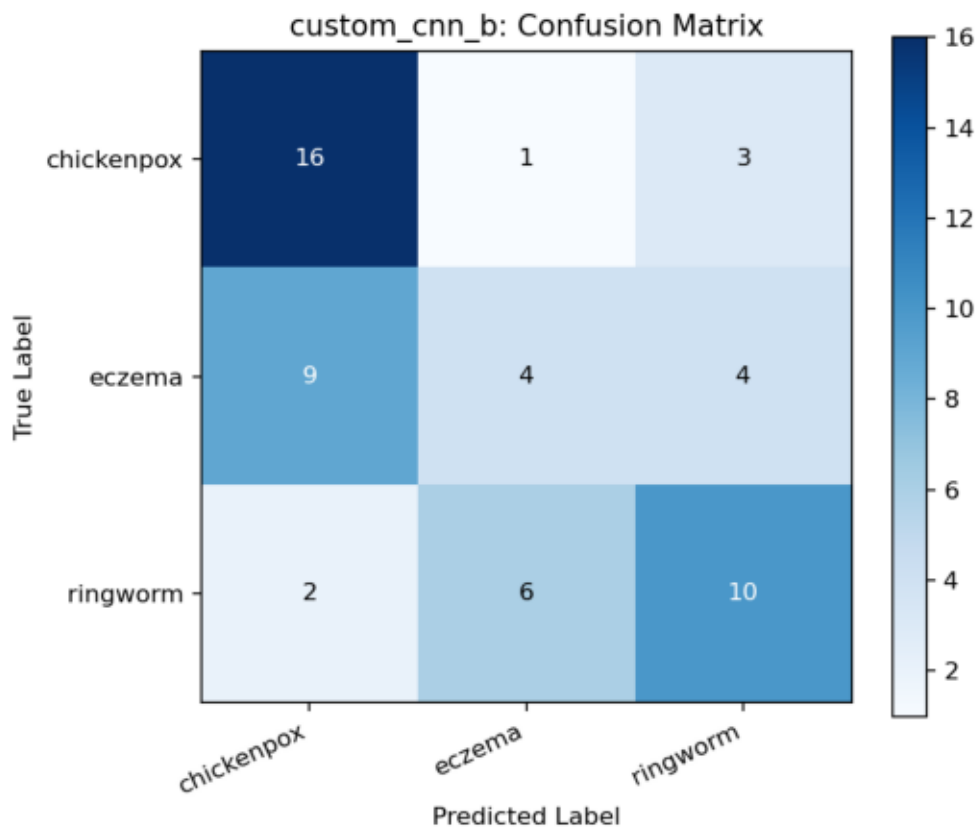
Results (Quantitative / Qualitative)

Skin Disease Detection CNN Project

Model A Confusion Matrix



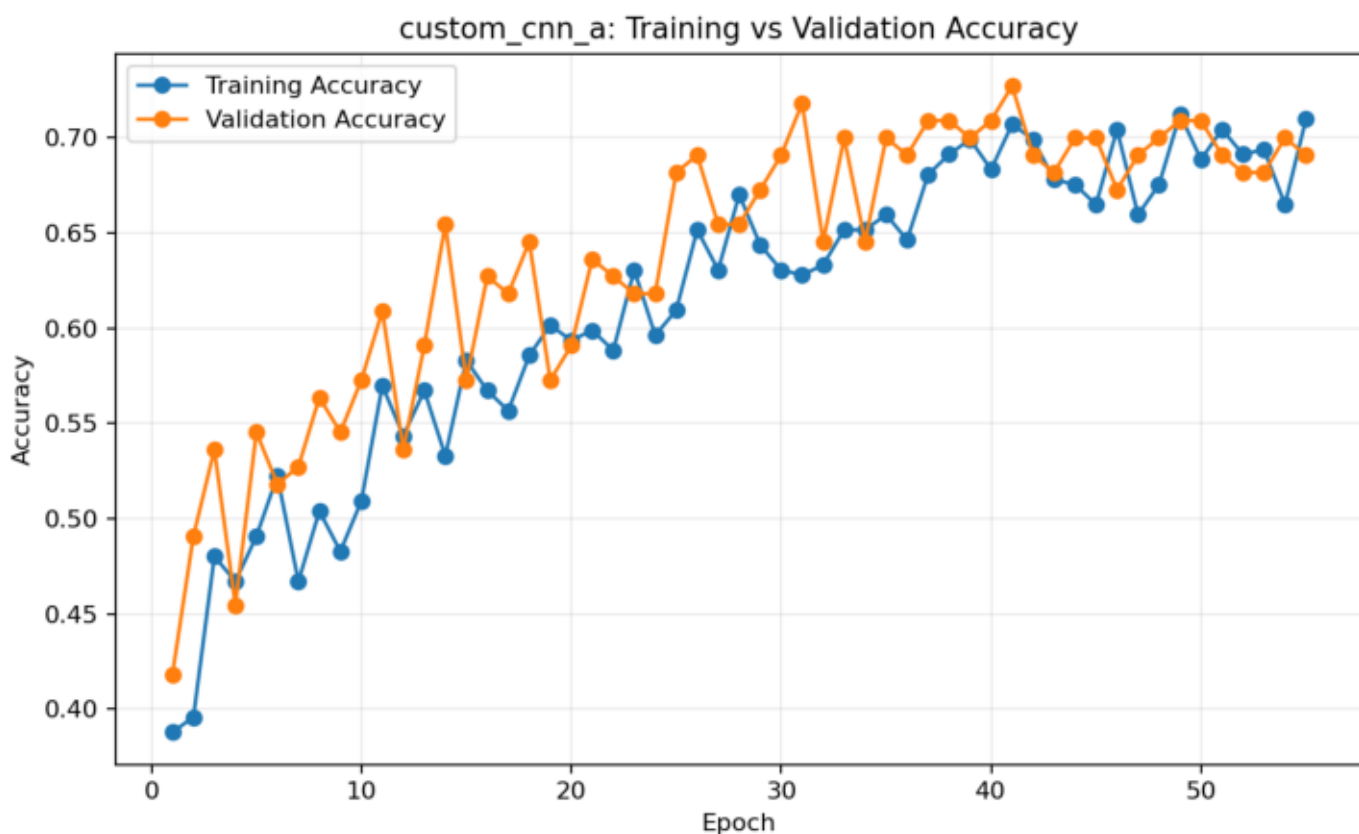
Model B Confusion Matrix



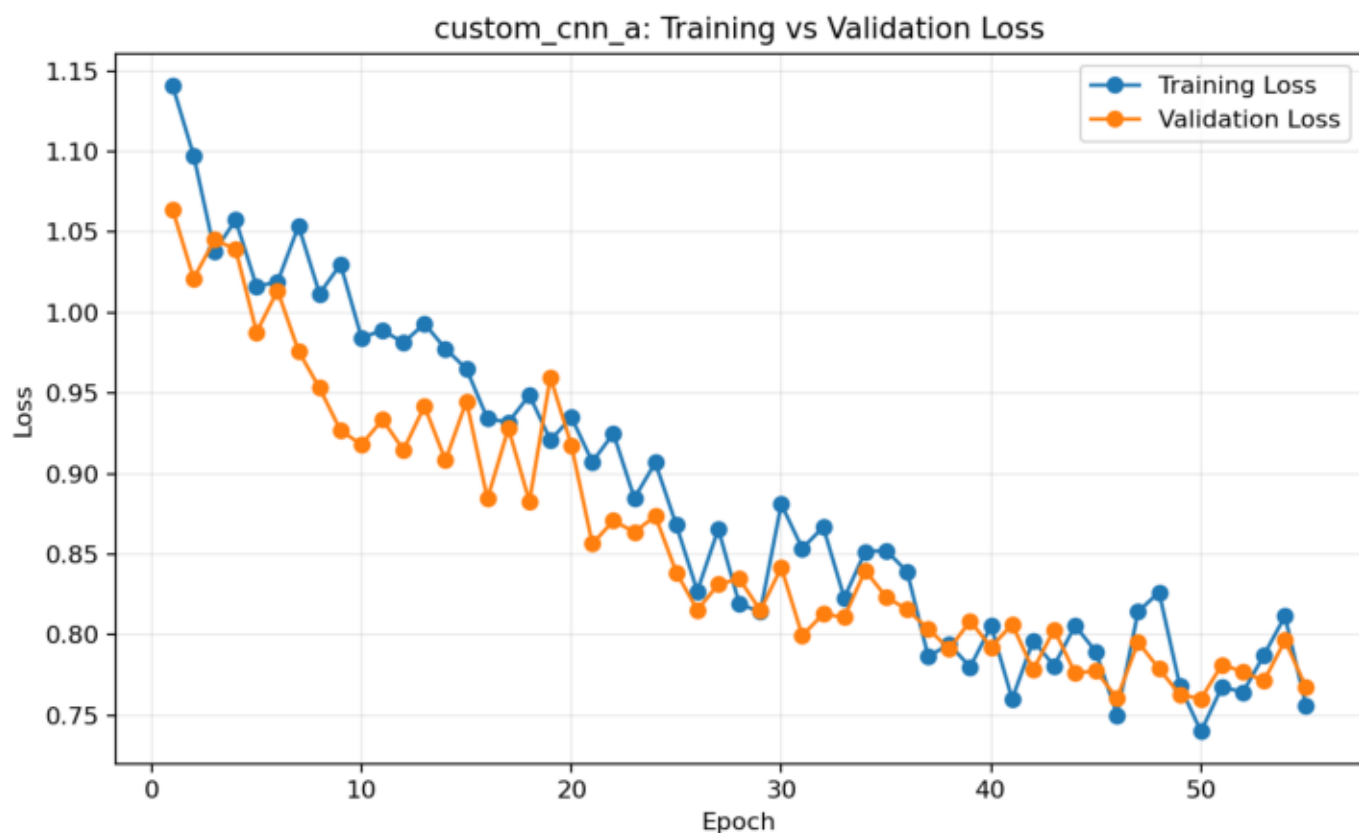
Results (Quantitative / Qualitative)

Skin Disease Detection CNN Project

Model A Accuracy Curve



Model A Loss Curve



Results (Quantitative / Qualitative)

Skin Disease Detection CNN Project

Model A Correct Prediction Samples

custom_cnn_a: Correct Predictions



Model A Incorrect Prediction Samples

custom_cnn_a: Incorrect Predictions



Conclusion

Skin Disease Detection CNN Project

Summary

Model A achieved the best latest test performance with 69.09% accuracy and 0.6831 macro F1-score. Model B achieved 54.55% accuracy and 0.5127 macro F1-score in its latest completed run.

Observation

The results show that the improved VGG-style CNN is more effective for the current dataset than the deeper plain CNN. Eczema remains the hardest class, indicating that more high-quality eczema samples or improved dataset cleaning would likely help more than simply increasing model complexity.

Future Work

Future improvements should focus on collecting more balanced images, removing mislabeled or low-quality samples, testing multiple random splits, and increasing data diversity while staying within the coursework restrictions.